A Comparative analysis of appearance based subspace methods for facial expression recognition

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***Abstract*—Facial expression recognition is like a pattern recognition problem, which is divided into two parts dimensionality reduction and classification. It is very much challenging to find out an efficient combination of dimensionality reduction technique and classifier. Many appearance based dimensionality reduction techniques have been proposed in recent years. This research focuses on the comparative analysis of appearance based subspace methods (PCA, KPCA, PCA+MI, KPCA+MI) for facial expression recognition in completely equal working conditions. PCA is a vastly used feature extraction technique which is linear and unsupervised. A kernelized version of PCA is called KPCA increases classification accuracy by handling non-linear data which is an unsupervised feature extraction method. KPCA shows the highest accuracy among these two methods. Combining a feature selection method MI with KPCA which can highly increase the recognition rate. The accuracy of KPCA+MI was above 95.24% when the moderate number of training features were used to train the SVM with RBF kernel.**

Keywords—Pattern Recognition; PCA; KPCA; MI; SVM

I. INTRODUCTION

Now a days, facial expression recognition technology has been able to attain more attention as interest is growing day by day in the field of expression information. It is a challenging and interesting problem which attracts researchers from different fields: psychology, pattern recognition, neural networks, and computer vision and computer graphics. In general, facial expression recognition techniques can be divided into two groups based on how the face is represented [1].

1) Appearance-based: It uses holistic texture features and is applied to either whole-face or specific regions in a face image.
2) Feature-based: It is based on the geometric relationship between facial features such as mouth, eyes, brows, cheeks, etc.

Appearance-based subspace analysis is one of the oldest and popular approaches which gives the most relevant results. Subspace analysis is performed by projecting an image into a space (subspace) which is lower dimension and after that recognition is performed by classifier on this space.

There are many feature extraction techniques for facial expression recognition with subspace analysis. For example LDA(Linear Discriminant Analysis)[2], PCA(Principal Component Analysis)[3], KPCA(Kernel Principal Component Analysis)[4], ICA(Independent Component Analysis)[5], PCA combined with LDA[6], etc. are the most extensively used feature extractions methods recommended by researchers.

This paper focuses on performance analysis of two appearance based subspace methods PCA and KPCA by following the same preprocessing step. How classification accuracy increased using a kernelized version of PCA is called KPCA by handling the non-linear relationship between pixels has been shown in the implementation section. This paper also shows a comparative analysis of two combined method PCA+MI, KPCA+MI with single PCA and KPCA methods. For classification purpose, Support Vector Machine has been used.

II. APPROACHES OF DIMENSIONALITY REDUCTION

Dimensionality reduction means reduce the number of features in feature space. A large number of feature makes the model more complex and creates the problem of overfitting. Dimensionality reduction can be implemented in two ways: feature extraction and feature selection. The main difference between feature selection and feature extraction is that feature extraction creates brand new features while feature selection keeps a subset of original features [7]. The Principal Component Analysis (PCA), Independent Component Analysis (ICA), Linear Discriminant Analysis (LDA), kernel-PCA are some feature extraction methods. Mutual Information (MI), minimum redundancy maximum relevance (mRMR) are some popular feature selection techniques.

A. Principal Component Analysis(PCA)

Principal Component Analysis (PCA) is a feature extraction technique that reduces the number of features in a large data set by transforming a large set of variables into smaller one by projecting them on a new space [8]. The number of features can be reduced without much loss of information.
PCA searches for k vectors which are orthogonal to each other. These vectors are n dimensional where $k \leq n$.These vectors are the basis for input data. As it does not consider the class label during feature extraction, it is an unsupervised feature extraction method. It finds the principal component on the basis of variance.

The steps involved in PCA are-
Step 1: Present the input data
Let $X = [x_1, x_2, x_3, .. x_n]$ and $Y = [y_1, y_2, y_3, ... y_n]$ are the input data.

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Step 2: Subtract the mean.
Calculation of mean,

$$X_m = \frac{\sum_{i=1}^{n} x_i}{n} \quad (1)$$

$$Ym = \frac{\sum_{i=1}^{n} y_i}{n} \quad (2)$$

The mean is subtracted from the dataset,

$$X = X - X_m \quad (3)$$

$$Y = Y - Y_m \quad (4)$$

Step 3: Calculate the covariance matrix.
Covariance is an estimate of the distribution of data in a data set. It helps to find out how much the dimensions vary from the mean with respect to each other. It is appraised of how much two random variables vary together.

$$cov(X,Y) = \frac{\sum_{i=0}^{n}(x_i - x_m)(y_i - y_m)}{n-1} \quad (5)$$

Where n defines the number of samples. Covariance is used to calculate the entries of the covariance matrix. The formula to find out the covariance matrix is

$$C = \begin{pmatrix} cov(x,x) & cov(x,y) \\ cov(y,x) & cov(y,y) \end{pmatrix} \quad (6)$$

.

Step 4: Calculation of the eigenvectors and eigenvalues of the covariance matrix
The direction of eigenvectors remains unchanged after applying a linear transformation on it. It can be expressed as

$$A\,v = \lambda v \quad (7)$$

Where v defines an eigenvector of A and λ is the corresponding eigenvalue.

Step 5: Choosing Principal Components
The eigenvalues have to be sorted in decreasing order. It serves as a new set of axes for the data which provides important information about the variance. Data size can be minimized by eliminating the components with low variance, without much loss of information. It should be possible to regenerate the original data using strong principal components.

B. Kernel Principal Component Analysis(KPCA)

KPCA is a feature extraction method that is non-linear. The Conventional PCA always identify a linear relationship between data point to represent the data in lower dimension space. But all real world data are not linear. The kernel PCA can identify the non-linear relationship of data. It is extended from PCA to represent nonlinear mappings in a high-dimensional feature space. Kernel PCA just performs PCA in a new space. Suppose that instead of using the original data points, they are mapped into a high dimensional feature space. Then the principal component is extracted from that space. However, it is computationally expensive to compute both the covariance matrix and its corresponding eigenvectors and eigenvalues in the high-dimensional feature space [9]. It is very difficult to compute the dot products of vectors with a high-dimension. It increases the computation time and cost. Kernel trick gives a solution to this problem. Using the kernel trick, all the calculations are performed in the original input space, but the advantages of high dimensional space can be achieved. In this way, an expensive mapping function can be avoided. In this research, KPCA using Gaussian kernel has been implemented. It can be expressed as:

$$\textbf{Gaussian}\ \ k(x_i, x_j) = e^{(-\gamma|x_i - x_j|^2)} \quad (8)$$

.

PCA finds a new dimension based on the covariance matrix of original variables but KPCA finds a new dimension based on the Kernel matrix. In KPCA, it does not possible to reconstruct the original image using few eigenvalues. As it does not consider the class label during feature extraction, it is an unsupervised dimensionality reduction technique.

Steps of finding the kernel of KPCA:

- Centered features:

$$\phi'(x_k) = \phi(x_i) - \frac{1}{n}\sum_{k=1}^{n}\phi(x_k) \quad (9)$$

- The corresponding kernel is:

$$k'(x_i, x_j) = \phi'(x_i)^T \phi'(x_j) \quad (10)$$

$$= \left(\phi(x_i) - \frac{1}{n}\sum_{k=1}^{n}\phi(x_k)\right)^T \left(\phi(x_J) - \frac{1}{n}\sum_{k=1}^{n}\phi(x_k)\right)^T$$
$$= k(x_i, x_j) - \frac{1}{n}\sum_{k=1}^{n} k(x_j, x_k) - \frac{1}{n}\sum_{k=1}^{n} k(x_j, x_k) + \frac{1}{n^2}\sum_{l.k=1}^{n} k(x_l, x_k) \quad (11)$$

- In a matrix form

$$k' = k - 2\mathbf{1}_{\frac{1}{n}}K + \mathbf{1}_{\frac{1}{n}}k\mathbf{1}_{\frac{1}{n}} \quad (12)$$

Where $\mathbf{1}_{\frac{1}{n}}$ is a matrix with all elements 1/n.

Steps of KPCA:
Step 1: Select a kernel.
Step 2: Construct the kernel matrix of the data

$$k' = k - 2\mathbf{1}_{\frac{1}{n}}K + \mathbf{1}_{\frac{1}{n}}k\mathbf{1}_{\frac{1}{n}} \quad (13)$$

Step 3: Solve an eigen value problem:

$$k'\alpha_i = \lambda_i \alpha_i \quad (14)$$

Step 4: New data in projected space

$$y_j = \sum_{i=1}^{n} \alpha_{ji} k(x, x_i), j = 1, \dots, d \quad (15)$$

C. Mutual Information(MI)

The amount of information that one random variable bears about another variable is called Mutual Information (MI) [10]. It is a measure of the dependency between two random variables say x and y. This dependency is based on the

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information shared between these two variables and if they are discrete, it is defined as:

$$I(x,y) = \sum_{i=1}^{n}\sum_{j=1}^{n} p(x(i),y(j))\log\left(\frac{p(x(i),y(j))}{p(x(i)).p(y(j))}\right) \quad (16)$$

Where P(x.y) is the joint probability function and P(x) and P(y) is the marginal probability function of x and y respectively.

By measuring how much each feature in a feature vector knows about classes, uncertainty about classes can be reduced [11]. Let *x* is the feature vector and *y* is the label of class where their combined uncertainty *H(x,y)* is less than individual uncertainty *H(x)* and *H(y)* of respectively.
The mutual information *I(x,y)* is defined as follows:

$$I(x,y) = H(x) + H(y) - H(x,y) \quad (17)$$

The relevant features are selected by ordering mutual information between each feature and classes in descending order such that I1>I2>I3…..>IN.

III. SUPPORT VECTOR MACHINE

SVM is a supervised learning algorithm which uses a labeled training sample to classify new observation by finding optimal decision boundary which maximizes the margin of training data. For training purpose, we can form a hyperplane which can separate two class accurately where all the points above the hyperplane belong to one class and all the points below the hyperplane belong to another class.

$$w_0 + w_1x_1 + w_2x_2 \geq 1 \;\; for\; y_i = +1 \quad (18)$$
$$w_0 + w_1x_1 + w_2x_2 \leq -1 \;\; for\; y_i = -1 \quad (19)$$

Here the point points fall on above the line belongs to positive class and the point fall on below the lines belongs to negative class [12]. The main challenge is to choose optimal separating hyperplane which can correctly classify real life data and generalize better with unseen data. There is already four different Support Vector Machines [13]**:**

1. The original one: The Maximal Margin Classifier.
2. The kernelized version using the Kernel Trick.
3. The soft-margin version.
4. The soft-margin kernelized version.

The last version is the combination of versions 1, 2 and 3. It is the most popular and widely used because it is able to classify non-linear datasets along with good margin. For solving non-linear problem different kernel is used. In this research purpose, Gaussian radial basis function kernel has been used. It is expressed as:

$$k(x_i, x_j) = e^{\left(\frac{-|x_i - x_j|^2}{2\sigma^2}\right)} \quad (20)$$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Input dataset

In this research purpose, we have used the JAFFE dataset. It is a widely used dataset for research purposes. JAFFE dataset is a balanced dataset. The size of each image is 256×256 Pixels, with 256 grey levels per pixels. The dataset contains 213 images of 7 facial expressions (angry, depressed, fear, happy, normal, sad, surprised) posed by 10 Japanese female models. So, the dataset has 7 classes. The images were captured in different light intensity. The database was designed and congregate by Michael Lyons, Miyuki Kamachi, and Jiro Gyoba. The images were captured at the Psychology Department at Kyushu University. [14].

B. Implementation

Among the total image, 70% has been used to train the classifier and 30% has been used to test the classifier.

The image is preprocessed by detecting face from the whole image. Face detection is performed by violazones [15] face detection algorithm. The algorithm finds out the facial portion and takes only the cropped portion of the face. The size of the original image is 256 × 256. But after face detection, the size of the image is 162×162.

As the research is based on subspace analysis of facial expression recognition algorithm, we have implemented two feature extraction methods PCA and KPCA. Feature selection method MI is used to select features from reduced space. Before implement any feature extraction method, every image has been converted into a column vector where each row is a feature. The image of N×N has been converted a column P_i vector with N^2 features. As every preprocessed image has sized 162 ×162, so it has been converted into a column vector with 26244 features.

For classification purposes, Support vector machine with Gaussian radial basis kernel has been used. It is a Gaussian function. As we have used Gaussian RBF kernel, we have to choose the value of parameter *C* and σ by parameter tuning. σ is the measure of standard deviation in the Gaussian distribution and parameter *C* defines the value of soft margin. It is a big challenge to select the proper value of *C* and σ parameter. 5-fold cross validation has been performed to find out the best combination of *C* and σ for our model. As our datasets is multiclass, so we have performed classification by one against all approaches, which train binary classification to discriminate one class from all others output the class with the largest prediction output of binary classification.

C. Result Analysis

Table I shows the accuracy comparison between PCA and KPCA by taking the different number of features. From the table, we can see, how accuracy differs by changing the number of features. For the equal number of features, KPCA outperforms then PCA. With 30 features, the recognition rate of PCA is 87.30% and KPCA is 88.89%. KPCA shows the highest accuracy of 92.86% with 60 features. There is a great

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impact of gamma parameter value on the performance of KPCA. So it has to be chosen carefully.

Table I: Accuracy comparison between PCA and KPCA

Number of features	PCA	KPCA	Gamma (KPCA)
10	77.78%	77.78%	8192
30	87.30%	88.89%	8192
50	87.39%	90.48%	4096
60	87.30%	92.86%	8192
70	84.13%	90.48%	16,384
90	84.13%	90.48%	8192
100	84.13%	90.48%	8192

From Table II, we can see how classification accuracy has been increased by using KPCA instead of PCA as a feature extraction method. The highest classification accuracy using PCA is 87.30% where for KPCA is 92.86%. From the table, we also can see that PCA with MI gives accuracy 92.06% with 30 features which is better than single PCA. KPCA with MI also outperforms than single KPCA. It shows 95.24% accuracy by selecting 30 features.

Table II: Accuracy comparison between various methods

Method	Accuracy	Cost(C)	Sigma(σ)
PCA	87.30%	32	8
PCA-MI	92.06%	16	8
KPCA	92.86%	64	16
KPCA-MI	95.24%	32	8

CONCLUSION

This paper presented an analysis of the two most popular appearance based subspace methods (PCA and KPCA) for facial expression recognition in an identical working environment to measure their performance. From the analysis, it can be seen that KPCA gives better performance than PCA by handling non-linear data. As KPCA is an unsupervised feature extraction method, it selects features based on variance. But all the time variance is not a good parameter for selecting features, it may lose some valuable information. To overcome this problem, a new method is proposed combining feature selection method MI with KPCA by solving the problem of non-linear dataset and selecting the features by measuring the similarity with ground truth. KPCA with MI gives 95.24% accuracy by selecting 30 features when Support Vector Machines with RBF kernel is used as a classifier.

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